Coffee Store REST API



A Spring Boot REST API for managing coffee products and stock.

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What does our application do?

1

Coffee Products

- View products
- Add new products
- Store product details

2

Stock Management

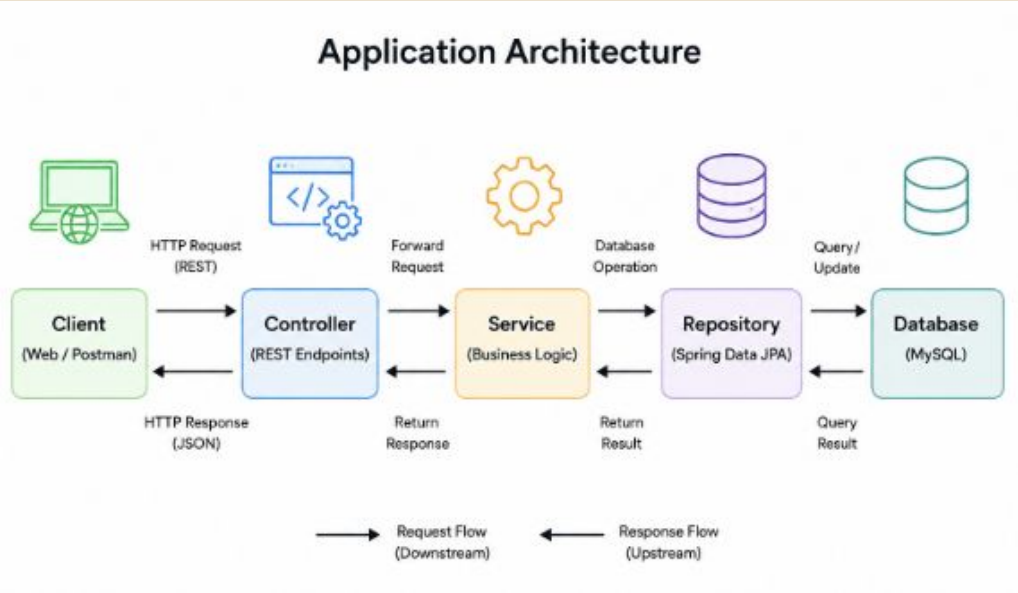
- Set stock quantities
- Increase/decrease stock
- Validate stock changes

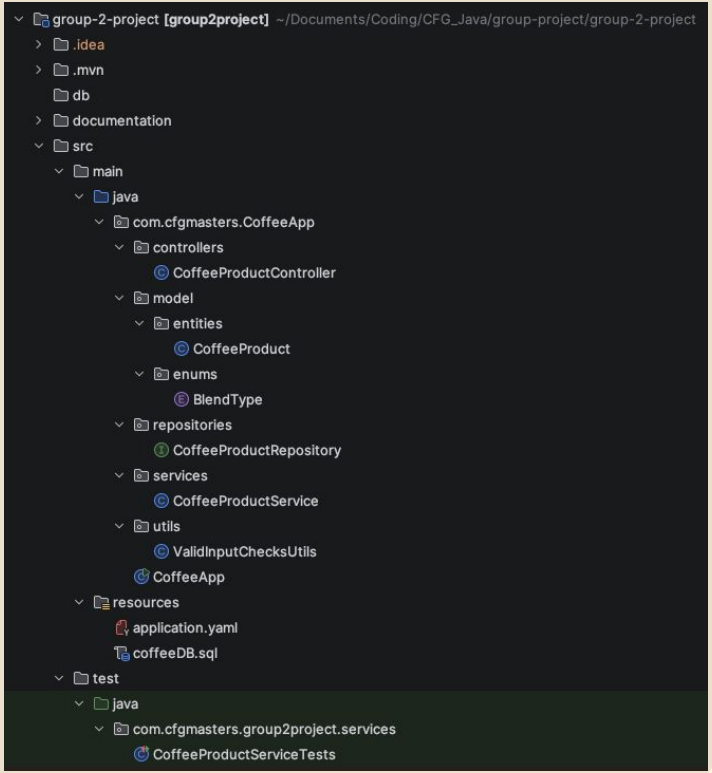
3

Database

- Products stored in MySQL
- Data accessed through Spring Data JPA

How we built it





Example: Stock Management

Our API provides two ways to manage product stock

Set stock to a specific quantity (20):

```
@Operation(summary = "Update coffee stock to a specific amount.")  Gen
@PutMapping("/stock/{id}")
public ResponseEntity<CoffeeProduct> updateStock(
        @PathVariable Long id,
        @RequestParam Integer quantity) {
    CoffeeProduct updatedProduct = coffeeProductService.updateStock(id, quantity);
    return ResponseEntity.ok(updatedProduct);
}
```

PUT /products/stock/1?quantity=20
- Sets the products stock to 20

Adjust existing stock:

```
@Operation(summary = "Adjusts coffee stock by an amount, can be negative or positive.")  Gen
@PutMapping("/adjustStock/{id}")
public ResponseEntity<CoffeeProduct> adjustStock(
        @PathVariable Long id,
        @RequestParam Integer amount) {
    CoffeeProduct updatedProduct = coffeeProductService.adjustStock(id, amount);
    return ResponseEntity.ok(updatedProduct);
}
```

PUT /products/adjustStock/1?amount=-5
- Reduces the existing stock by 5







Teamwork and Next Steps

1

Collaborative Development

- GitHub feature branches
- Individual feature development
- Merging into the main project

2

Code Integration

- Pull requests & code reviews
- Resolving merge conflicts
- Integrating team members' code

3

Shared Environment

- Docker & MySQL
- Consistent local setup
- Testing our integrated application

4

Next Steps

- Complete remaining functionality
- Finish testing with JUnit and Mockito
- Finalise documentation & Open API
- Final review before submission

5

Future Developments

- Complete remaining functionality
- Finish testing
- Finalise documentation & Open API
- Final review before submission